

Luke Yamaguchi

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EDUCATION

University of California, Los Angeles (UCLA)

Sep 2024 - Dec 2027

M.S. Electrical & Computer Engineering (Exceptional Student Admission Program)

Expected Dec 2027

B.S. Computer Engineering

Jun 2026

- **GPA:** 3.86/4.00

- **Relevant Coursework:** Computer Systems Architecture, Computer System Security, Digital Circuits, Operating Systems, Algorithms and Complexity, Data Communications and Telecommunication Networks, Communication Systems, Digital Signal Processing, Neural Networks and Deep Learning

EXPERIENCE

Graduate ASIC/FPGA Security Engineering Intern

Los Angeles, CA

The Aerospace Corporation

Jun 2026 - Sep 2026

- Building a test platform on the Xilinx Versal VEK280 to validate anti-tampering features
- Developing a DevSecOps pipeline for hardware trojan detection in ASIC/FPGA designs
- Prototyping an on-board LLM deployed on the VEK280 for real-time threat detection and autonomous mitigation

Undergraduate Researcher

Los Angeles, CA

Secure Systems and Architectures Lab - UCLA

Oct 2025 - Jun 2026

- Researched BLE hardware security and RF device authentication using PHY-layer hardware fingerprints
- Built GNU Radio DSP pipeline to extract physical-layer features from BLE signals captured from ADALM-Pluto SDR
- Trained ML-based RF fingerprinting models on Linux remote servers
- Developed universal adversarial perturbation attacks against RF fingerprinting models and conducted over-the-air spoofing experiments using HackRF SDR

PROJECTS

Hardware Security Lead — Mini-GPU FPGA Accelerator

Mar 2026 - Jun 2026

- Built a hardware root of trust for kernel verification with golden hashes, launch gating, and program-memory locking
- Threat-modeled zero-trust execution and patched CRC-failed writes, invalid ACKs, and residual memory after reset
- Cut SHA-256 core area 54% (10.5k to 4.9k cells) by serializing the digest adder and removing parallel logic
- Designed RTL UART datapath with RX/TX parsers, command decoding, CRC validation, and ACK/NACK signaling
- Closed timing at 75 MHz in Vivado with the secure design at 97.5% LUT utilization on Basys3 FPGA

Hardware-Enforced Authentication System

Feb 2025 - Mar 2025

- Designed and implemented multi-user authentication system in Verilog on an Artix-7 FPGA
- Built hardware-enforced role-based access control (RBAC) with admin, user, and guest privilege separation
- Implemented dynamic credential lifecycles, including creation, modification, and deletion
- Engineered fail-secure features (auto re-locking & brute-force lockout) informed by authentication threat modeling
- Verified authentication, credential management, and fail-secure functionality via simulation and .vcd waveform analysis

Lead Researcher — Multi-Agent Access Control

Oct 2025 - Dec 2025

- Proposed a provenance-based access control framework using instruction-origin tainting to mitigate Confused Deputy attacks in multi-agent LLM systems
- Reduced attack success rate from 57% to 29% across 15 test cases

SKILLS

Languages / HDLs: Verilog, SystemVerilog, C, C++, Python, Bash

Hardware / Architecture: RTL Design, FPGA Design, Versal SoC, Artix-7, ARM Cortex-M7, STM32

Verification/Synthesis: Testbenches, VCD Waveform Debugging, Vivado, Yosys, Icarus Verilog

Tools/Lab: Git, Linux, Docker, GDB, MATLAB, GNU Radio, Oscilloscopes, DMM, Analog Discovery 2

Languages: English, Japanese